Superactivation in two channel uses for private Horodecki channels with large shields: exact certificates

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Parentin, Bergh, Datta and Wilde made superactivation of quantum capacity a finite question. A PPT channel and a 50% erasure channel each transmit a qubit with channel fidelity at most 3/4 in any number of uses; the least number n^* of joint uses whose fidelity exceeds 3/4 measures the onset, and for Smith and Yard's pair they established that n^* is at most 17 with an explicit code whose fidelity they evaluated in floating point. We certify codes: a stored encoder and decoder whose fidelity is bounded below in exact or rigorous interval arithmetic, through a reduction we prove. For the Smith–Yard pair, a code supported on the symmetric subspaces of two groups of 7 and 8 uses, and not symmetric across them, has fidelity at least 0.7500784415, so n^* is at most 15. Smith and Yard's PPT channel is the smallest member of a family P_d of private Horodecki channels with shield dimension d, constructed by K. Horodecki *et al.*, which is PPT at one flip probability. We prove that P_d, for every complex Hadamard twist, is PPT for exactly that one value; that one joint use of P_d with a 50% erasure channel simulates a flagged qubit channel; that one use of any PPT channel with a 50% erasure channel never exceeds 3/4; and that two uses of P_d with a 50% erasure channel exceed 3/4 for every d from 20 on, by certificates at $d = 20$ and 21 and by the closed form $(7 - 6p + 2p^2)/8$ from $d = 22$ on. At threshold 3/4 the onset is therefore exactly two uses for every such d. For $d = 7$ to 21 we certify upper bounds on n^*: four uses at $d = 7$ and 8, and three for $d = 9$ to 19. At one use, a certified code on the joint channel beats every code on the reduced qubit channel through which all these codes are built, so that reduction is strictly lossy there. A PPT-assisted relaxation cannot certify $n^* > 2$: we prove its value at two uses is 7/8. Every result of ours above is proven or certified; no hardware demonstration is claimed.

I. INTRODUCTION

Smith and Yard showed that two channels with zero quantum capacity, a positive-partial-transpose (PPT) channel and a 50% erasure channel, can have positive quantum capacity when used together [1]. Their witness is single-letter: the coherent information of one joint use is at least half the private information of the PPT channel. In terms of rates, the effect is therefore visible at one use. What a rate does not say is how many joint uses a code needs before its fidelity exceeds anything either channel can reach alone. (Detecting positive capacity through the coherent information of n uses is a different question, with no bound on n valid for all channels [2].)

Parentin, Bergh, Datta and Wilde made this finite question precise [3]. They measure a code by its channel fidelity $F_c(\mathcal{N}, 2)$, the largest fidelity with which a qubit of a maximally entangled pair survives the channel, and show that neither of Smith and Yard's channels exceeds 3/4 in any number of uses (their Result 1). The pair is *n-shot superactivated* when n joint uses exceed 3/4, and the least such n is the onset n^*. Reducing each joint use to a flagged qubit channel $\widetilde{\mathcal{N}}$ (their Prop. 1) and optimizing permutation-invariant codes [4], they established $n^* \leq 17$ with an explicit code whose fidelity they evaluated in floating point (0.75013 in their text, 0.750131 in their ancillary data), and asked whether n^* can be lowered. Our independent composition of their stored code agrees with their stored data (Sec. V); the distinction we draw is between floating-point and rigorous evaluation, not doubt about their bound.

This paper certifies codes and uses the certificates to locate the onset (Table I):

1. *Certificates.* For a code given as stored bytes, we prove a lower bound on its fidelity: the rounded bytes define the encoder and decoder, positivity is proved exactly, normalization costs only a nonnegative completion, and the fidelity is evaluated in ball arithmetic or exactly in $\mathbb{Q}(\sqrt{d})$. For codes whose encoder output is supported on the symmetric subspaces of groups of uses, the evaluation runs through a reduction that we prove (Lemma 2) and also check, with the certifiers' own arithmetic, against the dense construction (Sec. III).

2. *The family.* Smith and Yard's PPT channel is the $d = 2$ member of the family P_d of private Horodecki channels with shield dimension d constructed by K. Horodecki, Pankowski, M. Horodecki and P. Horodecki [5], PPT at flip probability $p = 1/(\sqrt{d} + 1)$ (written out for the Fourier twist in Ref. [6]). We prove that P_d is PPT at no other p, for every complex Hadamard twist, that one joint use with a 50% erasure channel simulates a flagged qubit channel, that neither channel alone exceeds 3/4, and that one use of any PPT channel with a 50% erasure channel never exceeds 3/4 (Sec. IV). At $d = 2$ these are Ref. [3]'s Prop. 1 and Result 1, which they reproduce.

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3. *Smith and Yard's pair.* A code supported on the symmetric subspaces of two groups of uses, and not symmetric across them, certifies superactivation at $n = 15$ and $n = 16$, so $n^* \leq 15$ (Sec. V). That non-symmetric codes can beat symmetric ones is not new (Sec. V); what is new is a certified code below 17 uses.

4. *Two uses.* Two uses of P_d with a 50% erasure channel exceed 3/4 for every $d \geq 20$, so at threshold 3/4 the onset is exactly two uses (Sec. VI).

5. *The onset across the family.* Certified upper bounds on n^* for $d = 7, \ldots, 21$ (Sec. VII).

6. *The reduction is lossy at one use.* One use of the joint channel, coded directly, beats every code on $\widetilde{\mathcal{N}}$ (Sec. VIII).

7. *Limits.* We give no lower bound on n^* beyond $n^* \geq 2$, and show that a PPT-assisted relaxation cannot give one (Sec. IX).

Evidence classes. We label each statement. *Proven*: a proof is given here in full. *Certified*: a lower bound proved by exact or rigorous interval arithmetic from committed bytes, and re-derived by a committed checker. *Imported*: a result of Ref. [3], quoted and not re-proved; no statement of ours depends on one. *Derived*: a hand calculation, checked numerically, whose proof is not given in full. *Measured*: a floating-point computation, not certified. *Simulated*: a device model.

II. SETTING, AND THE TWO RESULTS OF REF. [3]

For a channel $\mathcal{N}$ from A to B, the channel fidelity for qubit transmission is

$$F_c(\mathcal{N}, 2) = \max_{\mathcal{E},\mathcal{D}} \langle \Phi | \big(\mathrm{id}_R \otimes \mathcal{D} \circ \mathcal{N} \circ \mathcal{E}\big)(|\Phi\rangle\langle\Phi|) |\Phi\rangle, \quad (1)$$

where $|\Phi\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ on a reference R and a qubit, and the maximum runs over every encoding channel $\mathcal{E}$ from the qubit to A and every decoding channel $\mathcal{D}$ from B to a qubit. For n uses the channel is $\mathcal{N}^{\otimes n}$. Any explicit encoder and decoder give a lower bound on F_c.

Definition 1 (Ref. [3], specialised to qubits)**.** *Channels P and A exhibit n-shot superactivation at threshold error $\varepsilon \in (0,1)$ if $\max\{F_c(P^{\otimes m}, 2), F_c(A^{\otimes m}, 2)\} \leq 1 - \varepsilon$ for every m and $F_c((P \otimes A)^{\otimes n}, 2) > 1 - \varepsilon$. The onset n^* is the least such n.*

Ref. [3] defines the channel fidelity and the definition for a message of any Schmidt rank; we use only rank 2, write $F_c(\cdot, 2)$, and reserve the letter d for the shield dimension of Sec. IV. We use $\varepsilon = 1/4$ throughout. Ref. [3] states:

Imported result 1 (Result 1 of Ref. [3])**.** *Let P be the private Horodecki channel and A the 50% erasure channel considered in Ref. [1]. Then $\max\{F_c(P^{\otimes m}, 2), F_c(A^{\otimes m}, 2)\} \leq 3/4$ for every m, and $F_c((P \otimes A)^{\otimes 17}, 2) > 3/4$.*

Their proof of the first part shows more for P: its outputs are PPT, so $F_c(P^{\otimes m}, 2) \leq 1/2$.

Imported result 2 (Prop. 1 of Ref. [3])**.** *There exist pre- and post-processing operations $(\widetilde{\mathcal{E}}, \widetilde{\mathcal{D}})$ for $P \otimes A$ such that $\widetilde{\mathcal{N}} = \widetilde{\mathcal{D}} \circ (P \otimes A) \circ \widetilde{\mathcal{E}}$ is the qubit-input channel*

$$\widetilde{\mathcal{N}}_p(\rho) = \tfrac{1}{2}\,\rho \otimes |0\rangle\langle 0|_Z + \tfrac{1}{2}\,(\mathcal{X}^p \circ \overline{\Delta})(\rho) \otimes |1\rangle\langle 1|_Z, \quad (2)$$

with $p = 1/(1+\sqrt{2})$, where $\mathcal{X}^p(\rho) = (1-p)\rho + pX\rho X$, $\overline{\Delta}$ is complete dephasing and Z is a classical flag.

Sec. IV proves both statements for every member of the family P_d, with $p = 1/(\sqrt{d}+1)$ (Lemmas 4 and 5), and so for $d = 2$, whose member is Smith and Yard's channel (Sec. IV). The supplementary material of Ref. [3] also states (Sec. S3.2) and proves (Secs. S3.3–S3.5) a general form of Prop. 1, for every channel whose Choi state is a mixture of shifted private states, of which Lemma 4 is an instance; we give the proof in full so that every step our results use is proven here.

Because $\widetilde{\mathcal{N}}$ is one use of $P \otimes A$ with processing before and after, data processing gives

$$F_c\big((P \otimes A)^{\otimes n}, 2\big) \geq F_c\big(\widetilde{\mathcal{N}}^{\otimes n}, 2\big) \quad \text{for every } n. \quad (3)$$

In $\widetilde{\mathcal{N}}^{\otimes n}$ the decoder reads the flags $z \in \{0,1\}^n$; each flagged qubit is classical, so it may be measured in Z without loss, and $F = \sum_z 2^{-n} F_z$ with F_z the fidelity of the branch with flags z.

III. CERTIFICATES AND THE GROUP REDUCTION

Floating-point see-saw optimizers [10] propose an encoder and decoders. A separate certifier then proves a lower bound for maps defined by the stored bytes; it never uses the proposer's numbers. Its steps:

The maps: Every stored entry is rounded to the dyadic grid 2^{-96} (for the symmetric and two-group certifiers, then shifted by 2^{-34} times the identity). The rounded matrices define the certified maps; nothing is approximated after this step.

Positivity: Exactly: either by construction, the encoder's Choi matrix being a Gram matrix $|\tilde{w}\rangle\langle\tilde{w}|$ and each decoder's $\tilde{K}\tilde{K}^\dagger$ for a rounded factor $\tilde{K}$, or by proving every block positive definite with Sylvester's criterion, each leading principal minor an exact integer determinant (Bareiss elimination for large blocks).

TABLE I. The results and their evidence classes. The proofs use standard results we cite: the fidelity bounds for PPT states [7] and for two-extendible states [8, 9], and data processing. At $d = 2$ the proven rows on the family are Prop. 1 and Result 1 of Ref. [3].

statement	class	where
The group reduction behind the symmetric and two-group certificates is exact	proven; also checked against the dense construction	Lemma 2
Smith–Yard pair ($d = 2$): a code with $n = 15$ uses has $F \geq 0.7500784415$, so $n^* \leq 15$ (published: $n^* \leq 17$)	certified; the steps from it to n^* proven	Thm. 7
P_d, the construction of Ref. [5], is PPT if and only if $p_1 = 1/(\sqrt{d}+1)$, for every complex Hadamard twist ("if" for a symmetric twist: Ref. [5]; "only if" for the Hadamard twist: its Obs. 2, and for every twist: here); P_2 is Smith and Yard's channel	proven; P_2 by exact computation	Thm. 3
One joint use of P_d and A_{2d} simulates $\widetilde{\mathcal{N}}_p$, $p = 1/(\sqrt{d}+1)$ (Smith and Yard's coupling; at $d = 2$, Prop. 1 of Ref. [3])	proven	Lemma 4
Neither P_d nor A_{2d} alone exceeds 3/4 at any number of uses	proven (standard argument)	Lemma 5
One use of any PPT channel with a 50% erasure channel: $F_c \leq 3/4$	proven	Lemma 6
Two uses of $P_d \otimes A_{2d}$: $F_c > 3/4$ for every $d \geq 20$	certified ($d = 20, 21$); proven ($d \geq 22$)	Thm. 8
At threshold 3/4, $n^* = 2$ exactly for every $d \geq 20$	follows from the rows above	Cor. 9
$n^* \leq 4$ for $d = 7, 8$ and $n^* \leq 3$ for $d = 9, \ldots, 19$	certified upper bounds	Table III
One use of the joint channel ($d = 2$): a code with $F \geq 0.6627576287$ beats every code on the reduced channel ($\max = 1 - \sqrt{2}/4 \approx 0.6464466094$)	certified and proven	Thm. 10
A PPT-assisted relaxation cannot certify a lower bound on n^* above 2: at n uses its value is exactly $1-2^{-(n+1)}$, for $\widetilde{\mathcal{N}}_p$, for A and for every pair $P_d \otimes A_{2d}$	proven; feasible point and dual certificate also checked exactly on the full output space for $\widetilde{\mathcal{N}}_p$ and A ($n \leq 3$) and for $P_2 \otimes A_4$ ($n = 1$)	Prop. 11

Normalization: Without square roots: an exact Gershgorin bound $\lambda_{\max} \leq 1 + e$ on the relevant partial trace makes the map divided by $1 + e$ trace non-increasing. Completing it to a channel adds a completely positive term whose contribution to the fidelity is non-negative, so dividing by the normalizers gives a lower bound for a pair of channels.

The fidelity: Either in 256-bit ball arithmetic, with p and every square root as a rigorous ball, or exactly: for $p = 1/(\sqrt{d}+1) = (\sqrt{d}-1)/(d-1)$ the dense certifier expands each flagged branch over the subsets of flipped bits, every term an exact non-negative integer at a fixed scale, so the bound is $a + b\sqrt{d}$ with a, b rational and $F > 3/4$ is decided by the sign of $(a - 3/4) + b\sqrt{d}$.

A wrong witness fails a positivity proof or certifies less. The certified number is a lower bound on $F_c(\widetilde{\mathcal{N}}^{\otimes n}, 2)$, which is a maximum over all codes.

The group reduction. Codes whose encoder output is supported on $R \otimes \mathrm{Sym}^{m_1} \otimes \cdots \otimes \mathrm{Sym}^{m_G}$, for consecutive groups of uses, are evaluated in a reduced representation; for $G = 1$ it is the symmetric-subspace case of the permutation-invariant evaluation of Refs. [3, 4]. (Invariance under permutations within groups is not enough: $\frac{1}{2}\mathbb{1}_R \otimes |\Psi^-\rangle\langle\Psi^-|$ is invariant under the swap of two qubits but orthogonal to Sym^2.) Let the n uses form consecutive groups of sizes $m_1, \ldots, m_G$ ($G = 1$ for symmetric codes, $G = 2$ for two-group codes), let Sym^m be the symmetric subspace of m qubits with Dicke basis $|D^m_w\rangle$ ($w = 0, \ldots, m$), and define $P^{(m)}_{k,a} \colon \mathrm{Sym}^m \to \mathrm{Sym}^{m-k}$ by

$$P^{(m)}_{k,a}|D^m_w\rangle = \left(\binom{m-k}{w-a} \Big/ \binom{m}{w}\right)^{1/2} |D^{m-k}_{w-a}\rangle, \qquad (4)$$

the term being absent when $w - a \notin [0, m-k]$. Let $\Pr_k(b|a)$ be the probability that k bits, a of them equal to 1, read b ones after each is flipped independently with probability p.

Lemma 2 (group reduction)**.** *Let ω be a state on $R \otimes (\mathbb{C}^2)^{\otimes n}$ supported on $R \otimes \mathrm{Sym}^{m_1} \otimes \cdots \otimes \mathrm{Sym}^{m_G}$, sent through $\widetilde{\mathcal{N}}_p^{\otimes n}$. Fix flags z with k_g flagged uses in group g, let Bob measure the flagged qubits in Z, and let b_g be the number of ones he reads in group g. After Bob relabels the qubits within each group so that the flagged ones come first, his operator on R and the unflagged qubits, summed over all outcome strings with weights $b = (b_g)$, is*

$$\rho_{k,b}(\omega) = \sum_a \prod_g \binom{k_g}{a_g} \Pr_{k_g}(b_g|a_g)\, Q_a\, \omega\, Q_a^\dagger, \qquad (5)$$

with $Q_a = \mathbb{1}_R \otimes \bigotimes_g P^{(m_g)}_{k_g,a_g}$, whatever the positions of the flags; it is supported on $R \otimes \bigotimes_g \mathrm{Sym}^{m_g - k_g}$. Consequently, for any decoding channels $\mathcal{D}_{k,b}$ on these registers, the decoder that reads the flags and the weights and

applies $\mathcal{D}_{k,b}$ achieves

$$F = \sum_k 2^{-n} \prod_g \binom{m_g}{k_g} \sum_b \langle \Phi | (\mathrm{id}_R \otimes \mathcal{D}_{k,b})(\rho_{k,b}(\omega)) | \Phi \rangle. \tag{6}$$

Proof. Splitting a Dicke state. The strings of weight w on m qubits split as (t, v), with t on the first k qubits and v of weight $w - |t|$ on the rest, and the uniform superposition of the v is $\binom{m-k}{w-|t|}^{1/2} |D^{m-k}_{w-|t|}\rangle$. Hence

$$|D^m_w\rangle = \sum_{t \in \{0,1\}^k} |t\rangle \otimes P^{(m)}_{k,|t|} |D^m_w\rangle,$$

so $(\langle t| \otimes \mathbb{1})|\psi\rangle = (\mathbb{1} \otimes P^{(m)}_{k,|t|})|\psi\rangle$ for every $|\psi\rangle \in \mathcal{H} \otimes \mathrm{Sym}^m$, with $\langle t|$ acting on the first k qubits of the group; the same holds group by group.

Placement. Let π permute qubits within groups and U_π act accordingly. U_π fixes $\mathrm{Sym}^{m_1} \otimes \cdots \otimes \mathrm{Sym}^{m_G}$ pointwise, so $U_\pi \omega U_\pi^\dagger = \omega$. The channel applied under flags $\pi(z)$ is $U_\pi \mathcal{N}_z (U_\pi^\dagger \cdot U_\pi) U_\pi^\dagger$, where $\mathcal{N}_z$ applies $\mathcal{X}^p \circ \overline{\Delta}$ to the flagged qubits and the identity elsewhere; so the output under $\pi(z)$ is $U_\pi \mathcal{N}_z(\omega) U_\pi^\dagger$, and Bob, who knows z, undoes U_π. It therefore suffices to flag the first k_g qubits of each group.

The flagged qubits. $\overline{\Delta}$ maps $|t\rangle\langle t'|$ to $\delta_{tt'}|t\rangle\langle t|$, and $\mathcal{X}^p$ flips a classical bit with probability p, so on the k flagged qubits and anything else,

$$\begin{aligned} &\big((\mathcal{X}^p \circ \overline{\Delta})^{\otimes k} \otimes \mathrm{id}\big)(\sigma) \\ &\qquad = \sum_s |s\rangle\langle s| \otimes \sum_t T(s|t)\, (\langle t| \otimes \mathbb{1})\sigma(|t\rangle \otimes \mathbb{1}), \end{aligned}$$

with $T(s|t) = \prod_i (1-p)^{[s_i = t_i]} p^{[s_i \neq t_i]}$. The flagged register is classical, so reading s loses nothing. By the splitting, $(\langle t| \otimes \mathbb{1})\omega(|t\rangle \otimes \mathbb{1}) = (\mathbb{1} \otimes \bigotimes_g P_{k_g,|t_g|})\, \omega\, (\cdot)^\dagger$.

Summing over strings. The sum of $T(s|t)$ over strings s with group weights b factorizes over groups, and in group g it is the probability that b_g ones are read from t_g; the flips are independent and identically distributed, so this depends on t_g only through $a_g = |t_g|$ and equals $\Pr_{k_g}(b_g | a_g)$. There are $\binom{k_g}{a_g}$ strings t_g of weight a_g, which gives (5); each P_{k_g,a_g} maps into $\mathrm{Sym}^{m_g - k_g}$, which gives the support.

The fidelity. The decoder of the statement is a channel: it reads z and s, relabels, and applies $\mathcal{D}_{k,b}$ through the isometry from the Dicke registers onto the symmetric subspaces of the unflagged qubits (completed arbitrarily outside them, where the state has no support). Each flag pattern has probability 2^{-n}, and $\prod_g \binom{m_g}{k_g}$ patterns have flag counts k; linearity in the outcome strings gives (6). □

The symmetric and two-group certifiers evaluate exactly the right side of (6) for the stored maps, in their own ball arithmetic, so with Lemma 2 their bounds are bounds on $F_c(\widetilde{\mathcal{N}}^{\otimes n}, 2)$; that arithmetic is checked against the dense construction below.

The reduction against the dense construction. We also checked Lemma 2 against the dense 2^n-dimensional construction, without using its formula: the encoder is embedded in the full space, the flagged qubits are permuted to the front, and for every flagged string t the slice that the reference and the unflagged qubits hold is computed directly, with the probability of each observed weight computed for that string t by convolution over its bits. For a random two-group code with groups of 5 and 7 uses ($n = 12$) every one of the 4,096 flag patterns was enumerated, with no symmetry assumed; for the certified two-group codes at $n = 4, 8, 15, 16$ and the certified symmetric code at $n = 17$, every flag-count branch was checked at the first positions and at random positions inside each group, with the stored decoders. In every case the slices satisfy the splitting identity, nothing leaves the symmetric subspaces, the branch states equal $\rho_{k,b}$, and the fidelity computed from the dense states equals the reduction's (for the certified codes, the recorded value of the stored pair), all to 7.8×10^{-15} (fidelities to 3.4×10^{-16}). For the symmetric certificates at $n = 20$ and 24, whose embedding we do not form, the slices of each Dicke state are counted over all 2^n strings, for every flagged string at both placements; this checks the splitting identity and the support, not the states.

The comparison also covers the certifiers' own ball arithmetic, which is code separate from the floating-point reduction above. The midpoints of the selection matrices the certificates multiply by equal the dense slices for every flagged string at every certified n, and the midpoints of their coefficients equal the per-string law summed over the strings of each weight for every group size up to 24 (to 2.0×10^{-15} per string), every ball radius (per string, for the coefficients) being at most 1.7×10^{-76}. Run now on every witness except the two-group ones at $n = 15, 16$, whose certificates take hours (two-group $n = 4, 8$; symmetric $n = 17, 20, 24$), each certifier reproduces its committed bound digit for digit, and the fidelity of the same stored pair (from the dense states, and as recorded at $n = 20, 24$) lies above that bound by less than 7.88×10^{-10} (the largest gap, at $n = 24$; Table II rounds the one and truncates the other to ten digits). Four negative controls each change one quantity by a relative 10^{-6}, and each is caught far above the deviations just quoted. Changing one amplitude of the reduction's selection matrix in the comparator's expectation (deviation 4.4×10^{-7}) shows that the identity check can fail. Changing the same amplitude inside the module that produces the reduced states (8.4×10^{-9}) shows that the state comparison runs the code the stored pairs were evaluated with, not a copy of it. Changing p in the dense per-string law (1.7×10^{-8}) shows that the comparison sees the per-string probabilities, where an error in the symmetry argument would appear. Changing one midpoint of a certifier's selection matrix (5.4×10^{-7}) shows that the certifier comparison reads the certifier's own matrices.

Four certifiers. *Symmetric codes* ($G = 1$) and *two-group codes* ($G = 2$), through Lemma 2, in ball arithmetic. *The n-use certifier*, for any code in the dense model, exact in $\mathbb{Q}(\sqrt{d})$, used for $n \le 4$. At $n = 2$ two further checks: a ball-arithmetic two-use certifier, and an exact-rational checker that imports none of our code, with explicit physical decoders (encoder inversion, single-qubit corrections given by integer quaternions, measure-and-prepare) and p enclosed by integer square roots. On a common code the n-use and two-use certifiers agree (a test holds them together).

Binding. Each certificate records the SHA-256 of the bytes it proved and the git objects of the code that proved it. The test suite recomputes each stored pair's floating-point fidelity and requires the certified bound to sit just below it: a bound above its own pair certifies some other pair. A check of this kind retracted the first record of the $n = 15$ certificate, which had proved decoders held only in memory; the result below was re-proved from stored decoders.

IV. THE FAMILY P_d AND WHAT ONE USE GIVES

Shifted private states. Let the key systems R', B' be qubits and the shields R'', B'' be d-dimensional. For $\ell \in \{0, 1\}$ let

$$\gamma^\ell = \frac{1}{2} \sum_{i,j \in \{0,1\}} |i\rangle\langle j|_{R'} \otimes |i \oplus \ell\rangle\langle j \oplus \ell|_{B'} \otimes \tau^\ell_{ij}, \qquad (7)$$

where $\tau^\ell_{ij} = U^{i,\ell} \sigma^\ell U^{j,\ell\dagger}$, with shield states $\sigma^0 = \mathbb{1}/d^2$ and $\sigma^1 = \frac{1}{d}\sum_a |aa\rangle\langle aa|$ on $R''B''$, and twists $U^{0,\ell} = \mathbb{1}$, $U^{1,0} = D_u F$ and $U^{1,1}$, where F swaps R'' and B'', $D_u = \sum_{a,b} u_{ab}|ab\rangle\langle ab|$, and $U^{1,1}|aa\rangle = d^{-1/2}\sum_b u_{ab}|bb\rangle$ on $\mathrm{span}\{|aa\rangle\}$ and $U^{1,1}$ is the identity on its orthogonal complement. Here u is a complex Hadamard matrix: $|u_{ab}| = 1$ and $uu^\dagger = d\,\mathbb{1}$ (for example the Fourier matrix), so that D_u, $U^{1,1}$ and $u/\sqrt{d}$ are unitary. Let P_d be the channel on $\mathbb{C}^{2d}$ with Choi state

$$\Phi^{P_d} = p_0\gamma^0 + p_1\gamma^1, \qquad p_0 + p_1 = 1; \qquad (8)$$

its marginal on the input $R'R''$ is maximally mixed (each τ^ℓ_{ii} has $\mathrm{Tr}_{B''}\,\tau^\ell_{ii} = \mathbb{1}/d$), so it is a channel.

P_2 is Smith and Yard's channel. For $d = 2$, $u = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ and $p_1 = 1/(\sqrt{2}+1)$, the Choi matrix (8) equals, entry by entry and exactly, that of Smith and Yard's channel as Ref. [3] states it: its six Kraus operators (Eqs. S2.20–S2.25 of its supplement, with M_0 and M_1 from Eqs. S2.26–S2.27) and its Choi state (Eqs. S2.4–S2.17). Against both, each of the 256 entry differences is an algebraic number whose minimal polynomial is x, and so are those between the Kraus operators and the Choi-state formula; in floating point the differences are at most 7.0×10^{-17}. No local unitary is needed. The check can fail: with the sign error in M_1 that the supplement reports in Smith and Yard's original Kraus operators, 8 entries differ (by up to 0.0733), and with the formula's weights q_0, q_1 and q_2, q_3 exchanged, 28 do.

Origin. The family is the construction of K. Horodecki, Pankowski, M. Horodecki and P. Horodecki [5], written out for the Fourier matrix by Badziąg *et al.* [6] (their Appendix A). In the X-form of a private bit, γ^0 has $X \propto W_{u^\dagger}$ and γ^1 has $X \propto W^\Gamma_{\bar{u}}$, with $W_U = \sum_{ij} U_{ij}|ij\rangle\langle ji|$ and Γ the partial transpose on B''; theirs are W_U and W^Γ_U for one unitary U with $|U_{ij}| = 1/\sqrt{d}$. So for a symmetric u (the Hadamard, Sylvester and Fourier matrices) Eq. (8) is their state with $U = \bar{u}/\sqrt{d}$. For a non-symmetric u the two terms carry the twists $u^\dagger$ and $\bar{u}$; when $\bar{u}_{ij}/\bar{u}_{ji} = r_i/r_j$ for some phases r_i, the product $\mathrm{diag}(\sqrt{r_i})$ on R'' and its conjugate on B'' multiplies the entries of $W_{u^\dagger}$ by r_i/r_j and fixes $W^\Gamma_{\bar{u}}$, so it maps the state onto theirs with $U = \bar{u}/\sqrt{d}$; whether every non-symmetric u is locally equivalent to one of their states we have not settled (a test checks both cases). They show that their state is invariant under partial transposition ("PPT invariant") at the weights fixed by the trace norms of W_U and W^Γ_U, which put our weight $p_1 = 1/(\sqrt{d}+1)$ on the W^Γ_U term when $|U_{ij}| = 1/\sqrt{d}$; that for U the Hadamard gate every other weight gives a state that is not PPT; and that the same argument shows their state with $X = W_U$ to be extremal among PPT states whenever X is Hermitian and X or X^Γ has a positive eigenvalue (their Observation 2 and the argument after it). Smith and Yard's channel is their $d = 2$ state with $U = H$ [1]. The next theorem proves both directions for every complex Hadamard u; what it adds to theirs is the "only if" for every such twist, stated as such, in particular when $W_{\bar{u}}$ is not Hermitian (the Fourier matrix for $d \ge 3$, for example).

Theorem 3 (the PPT point of Ref. [5], and its uniqueness)**.** *For every $d \ge 2$ and every complex Hadamard matrix u, the state* (8) *has positive partial transpose if and only if $p_1 = 1/(\sqrt{d}+1)$. In that case P_d is a PPT channel.*

Proof. Transpose $B'B''$. An off-diagonal key term $|i\rangle\langle j|_{R'} \otimes |i\oplus\ell\rangle\langle j\oplus\ell|_{B'}$ $(i \ne j)$ becomes $|i\rangle\langle j| \otimes |j\oplus\ell\rangle\langle i\oplus\ell|$, so the partial transpose is block diagonal in the key sectors $S_0 = \mathrm{span}\{|00\rangle, |11\rangle\}$ and $S_1 = \mathrm{span}\{|01\rangle, |10\rangle\}$ of $R'B'$. In S_0 the diagonal blocks come from γ^0 and the off-diagonal ones from γ^1; in S_1 it is the reverse. Two facts fix the diagonal blocks: $\sigma^0 \propto \mathbb{1}$ is invariant under unitaries and transposition, and $U^{1,1}\sigma^1 U^{1,1\dagger} = \sigma^1$ because $U^{1,1}$ acts on $\mathrm{span}\{|aa\rangle\}$, where $\sigma^1 = \mathbb{1}/d$, as the unitary $u^{\mathsf{T}}/\sqrt{d}$.

Sector S_0. Since $\sigma^1 U^{1,1\dagger} = d^{-3/2}\sum_{a,b} \bar{u}_{ab}|aa\rangle\langle bb|$, whose partial transpose on B'' is $d^{-3/2}M$ with $M = \sum_{a,b} \bar{u}_{ab}|ab\rangle\langle ba|$, the block is

$$\begin{pmatrix} \alpha\mathbb{1} & \beta M \\ \beta M^\dagger & \alpha\mathbb{1} \end{pmatrix}, \qquad \alpha = \frac{p_0}{2d^2}, \quad \beta = \frac{p_1}{2d^{3/2}}.$$

$M^\dagger M = \sum_{a,b} |u_{ab}|^2 |ba\rangle\langle ba| = \mathbb{1}$, so M is unitary and the eigenvalues are $\alpha \pm \beta$.

Sector S_1. The off-diagonal block is $\frac{p_0}{2d^2} N$ with N the partial transpose of $(D_u F)^\dagger = \sum_{a,b} \bar{u}_{ba} |ab\rangle\langle ba|$, that is $N = \sum_{a,b} \bar{u}_{ba} |aa\rangle\langle bb|$, and the diagonal blocks are $\frac{p_1}{2}\sigma^1$. Both act within $\mathrm{span}\{|aa\rangle\}$, where $\sigma^1 = \mathbb{1}/d$ and N is the matrix $u^\dagger$, with $NN^\dagger = d\,\mathbb{1}$; they vanish on its complement. The nonzero eigenvalues are therefore $\frac{p_1}{2d} \pm \frac{p_0}{2d^{3/2}}$.

All eigenvalues are non-negative if and only if $p_0 \geq \sqrt{d}\, p_1$ (sector S_0) and $\sqrt{d}\, p_1 \geq p_0$ (sector S_1), that is $p_0 = \sqrt{d}\, p_1$, or $p_1 = 1/(\sqrt{d}+1)$. A channel whose Choi state is PPT maps every state of a reference and its input to a state that is PPT across the reference and its output. □

The test suite checks the predicted spectrum numerically on both sides of the boundary and the PPT property at the boundary for Fourier and real Hadamard twists; just below the boundary the partial transpose has a negative eigenvalue (a negative control).

From now on $p = p_1 = 1/(\sqrt{d}+1)$, and A_{2d} is the 50% erasure channel on $\mathbb{C}^{2d}$. We write the input of P_d as a key qubit K_1 and a shield S_1, that of A_{2d} as $K_2 S_2$, and the output of P_d as $B'B''$. The coupling below is Smith and Yard's, who send a purification of the PPT channel's input through the erasure channel [1]; the decoder is that of Ref. [3] (their Prop. 1 at $d = 2$, and the general form of their supplement).

Lemma 4 (one joint use simulates $\widetilde{\mathcal{N}}_p$)**.** *With the encoder $V|r\rangle = d^{-1/2} \sum_s |r,s\rangle_{K_1S_1} \otimes |r,s\rangle_{K_2S_2}$ and the decoder below, one use of $P_d \otimes A_{2d}$ simulates exactly the channel $\widetilde{\mathcal{N}}_p$ of* (2)*, with A_{2d}'s erasure flag as the flag Z. Hence $F_c((P_d \otimes A_{2d})^{\otimes n}, 2) \geq F_c(\widetilde{\mathcal{N}}_p^{\otimes n}, 2)$ for every n.*

Proof. With the normalized Choi state, $P_d(Y) = 2d\ \mathrm{Tr}_{R'R''}[(Y^\mathsf{T} \otimes \mathbb{1})\Phi^{P_d}]$. For $Y = |r,s\rangle\langle r',s'|$ only the terms $i = r$, $j = r'$ of (7) survive:

$$P_d\big(|r,s\rangle\langle r',s'|\big) = d \sum_\ell p_\ell\, |r \oplus \ell\rangle\langle r' \oplus \ell|_{B'} \otimes \langle s|\tau^\ell_{rr'}|s'\rangle_{R''},$$

an operator on $B'B''$. The encoder gives $V|r\rangle\langle r'|V^\dagger = \frac{1}{d}\sum_{s,s'} |r,s\rangle\langle r',s'| \otimes |r,s\rangle\langle r',s'|$.

A delivers (probability 1/2). Since $\sum_{s,s'} |s\rangle\langle s'|_{S_2} \otimes \langle s|\tau|s'\rangle_{R''}$ is τ with R'' renamed S_2, the output is

$$\sum_\ell p_\ell\, |r \oplus \ell, r\rangle\langle r' \oplus \ell, r'|_{B'K_2} \otimes U^{r,\ell}\sigma^\ell U^{r',\ell\dagger} \quad \text{on } S_2B''.$$

Bob applies a CNOT from K_2 to B', which maps $|r \oplus \ell, r\rangle$ to $|\ell, r\rangle$, and measures B', obtaining ℓ with probability p_ℓ; the same ℓ appears in the ket and the bra, so the coherence between r and r' is kept. He then applies $\sum_i |i\rangle\langle i|_{K_2} \otimes U^{i,\ell\dagger}$ on S_2B'', which maps $|r\rangle\langle r'| \otimes U^{r,\ell}\sigma^\ell U^{r',\ell\dagger}$ to $|r\rangle\langle r'| \otimes \sigma^\ell$, and outputs K_2: the result is $|r\rangle\langle r'|$, the identity channel.

A erases (probability 1/2). A's output is the flag whatever its input, so the trace over K_2S_2 leaves $\frac{1}{d}\sum_s \delta_{rr'} P_d(|r,s\rangle\langle r,s|) = \delta_{rr'} \sum_\ell p_\ell |r \oplus \ell\rangle\langle r \oplus \ell|_{B'} \otimes \mathrm{Tr}_{R''}\, \tau^\ell_{rr}$. Bob discards B'' and outputs B': $\delta_{rr'} \sum_\ell p_\ell |r \oplus \ell\rangle\langle r \oplus \ell|$, which is $(\mathcal{X}^{p_1} \circ \overline{\Delta})(|r\rangle\langle r'|)$.

Bob reads the flag, which is orthogonal to the rest of A's output, applies the matching branch and keeps the flag; every step is a channel, so the composition is $\widetilde{\mathcal{N}}_{p_1}$. Applying V and the decoder to each use turns any code for $\widetilde{\mathcal{N}}_p^{\otimes n}$ into a code for $(P_d \otimes A_{2d})^{\otimes n}$ with the same fidelity. □

At $d = 2$ this is Imported result 2, since P_2 is Smith and Yard's channel. The proof uses only that the Choi state is a mixture of shifted private states. Tests also check both branches numerically for small d (Knill–Laflamme conditions; Helstrom error p), with a negative control.

Lemma 5 (neither channel alone)**.** $F_c(P_d^{\otimes m}, 2) \leq 1/2$ *and* $F_c(A_{2d}^{\otimes m}, 2) \leq 3/4$ *for every m and d.*

Proof. The standard argument, as in Ref. [3]. *P_d.* By Theorem 3 the Choi state of P_d is PPT, so is that of $P_d^{\otimes m}$ (a tensor product of PPT states), and $P_d^{\otimes m}$ maps every input to a state that is PPT across reference and output. The encoder acts before the channel and the decoder locally after it, so the final state on R and Bob's qubit is PPT, and its fidelity with Φ is at most 1/2 [7]. *A_{2d}.* A_{2d} has the isometric extension $|\psi\rangle \mapsto (|\psi\rangle_B|e\rangle_E + |e\rangle_B|\psi\rangle_E)/\sqrt{2}$, symmetric under exchanging B and E. For m uses and any encoder, the joint state of R, B^m and E^m is therefore symmetric under exchanging B^m and E^m; applying the decoder to B^m and a copy of it to E^m gives a state of R and two qubits, symmetric in the two qubits, whose marginal on R and the first qubit is the code's output. That output is two-extendible, and a two-extendible state has fidelity at most 3/4 with Φ [8, 9]. (The same bound holds when the uses are interleaved with two-extendible channels [11].) □

At $d = 2$ this gives the first part of Imported result 1.

Lemma 6 (one use never suffices)**.** *For any PPT channel P and any 50% erasure channel A, $F_c(P \otimes A, 2) \leq 3/4$.*

Proof. A's flag is classical and orthogonal to its other outputs, so the decoder may read it first, and $F = \frac{1}{2}F_{\mathrm{kept}} + \frac{1}{2}F_{\mathrm{erased}}$ for the fidelities conditioned on the flag. $F_{\mathrm{kept}} \leq 1$. When A erases, its output carries no information about its input, and the map from the transmitted qubit to Bob's output is a channel applied before P, then P, then the decoder. Its output on half of Φ is PPT across R and Bob's qubit, because P's outputs are PPT across reference and output and the decoder acts on the output alone, so $F_{\mathrm{erased}} \leq 1/2$ [7]. Hence $F \leq 3/4$. □

V. SMITH AND YARD'S PAIR: $n^* \leq 15$

Every code that Ref. [3] reports near the onset is permutation-invariant. A two-group code has its encoder output supported on $R \otimes \mathrm{Sym}^a \otimes \mathrm{Sym}^b$ with $a + b = n$,

so it is invariant under permutations within each group of uses, but not across them. Bergh and Parentin note that their see-saw can be restricted to a subgroup when the channel is covariant only under it: for $\mathcal{N}_1^{\otimes k} \otimes \mathcal{N}_2^{\otimes n-k}$ "one can restrict to encoders and decoders that are invariant under the action of the subgroup $G = S_k \times S_{n-k}$", and "A similar case arises for the channels involved in superactivation" (Ref. [4], the Remark after Lemma IV.9 in Sec. IV.C and its footnote 3); there the decoders of Ref. [3] for k flagged uses are covariant under $S_k \times S_{n-k}$ (Sec. S5.2.3 of its supplement). The channel $\widetilde{\mathcal{N}}^{\otimes n}$ itself is covariant under all of S_n, so restricting the encoder to two groups is a choice of code family, not a symmetry of the channel; what is new is a certified code of that family, evaluated through Lemma 2. The family contains every symmetric code ($a = 0$) and, at $n = 2$, every code. We grew two-group codes from smaller n by a see-saw, seeding each size with the code found one size below.

Theorem 7. *There is a code for $\widetilde{\mathcal{N}}^{\otimes 15}$ with a two-group encoder ($a = 7$, $b = 8$) and $F \geq 0.7500784415$, and one for $\widetilde{\mathcal{N}}^{\otimes 16}$ ($a = 8$, $b = 8$) with $F \geq 0.7503771797$. Smith and Yard's channels therefore exhibit 15-shot superactivation at threshold 3/4, and $n^* \leq 15$.*

Proof. The two bounds are certified through Lemma 2. By Lemma 4 at $d = 2$, since P_2 is Smith and Yard's channel (Sec. IV), $F_c((P \otimes A)^{\otimes 15}, 2) \geq F_c(\widetilde{\mathcal{N}}^{\otimes 15}, 2) > 3/4$; by Lemma 5 neither channel alone exceeds 3/4 at any number of uses. These two steps are also Imported results 2 and 1. □

Certified. The $n = 15$ certificate proves 1,621 blocks positive definite in exact integer arithmetic and took 2.1 h; its margin over 3/4 is at least 7.8×10^{-5}. At $n = 16$: 2,026 blocks, 4.6 h, margin at least 3.7×10^{-4}, so the crossing holds at two consecutive n and the margin grows. Table II sets these beside the symmetric codes. At $n = 14$ the two-group search (groups of 7 and 7) stalled at 0.7497607, below 3/4; whether some code with $n \leq 14$ crosses is open.

What is and is not new. Ref. [3] itself plots, in its Fig. 2, codes found by an unrestricted see-saw above its symmetric codes at every $n > 1$ for which it shows both (its text does not discuss the comparison). That symmetric codes are not optimal is therefore not new. What is new is a certified code, of any kind, below 17 uses.

Symmetric codes past the published range. Our symmetric codes for $n = 17, \ldots, 24$, scored with exact decoders, rise monotonically (Table II); Fig. 2 of Ref. [3] stops short of the upper end of this range. Three are certified: $F \geq 0.7502033333$ at $n = 17$, $F \geq 0.7506549819$ at $n = 20$ and $F \geq 0.7510307324$ at $n = 24$, a margin of at least 1.0×10^{-3} (326 blocks proved positive definite). Each value is an achievable code, not a converged optimum: every search stopped at its step limit. The encoders found are numerically isometries: truncating an encoder to its top eigenvector does not lower its fidelity (measured at every n; a test holds it at $n = 24$).

TABLE II. Codes for Smith and Yard's pair ($d = 2$). *Code*: a+b for a two-group code, sym. for a symmetric one. *Authors'*: the floating-point values of symmetric (isometric-ansatz) codes in the public table of Ref. [3]'s authors (`https://github.com/bbbergh/permqit`, commit `00c18d3d64`); the published paper prints 0.75013 for its code at $n = 17$ (0.750131 in its ancillary data), and the table's value at $n = 17$ belongs to a later code of the same ansatz, better by 4.1×10^{-5}. *F (float)*: our code's fidelity with its stored decoders (two-group) or with exact decoders (symmetric). *Certified*: the lower bound, truncated. The $n = 14$ row is the best code our search found there; it bounds nothing about other codes.

n	code	authors'	F (float)	certified $F \geq$
14	7+7	0.7492684	0.7497607	–
15	7+8	0.7496056	0.7500784437	0.7500784415
16	8+8	0.7499206	0.7503771821	0.7503771797
17	sym.	0.7501717	0.7502033339	0.7502033333
18	sym.	–	0.7503786523	–
19	sym.	–	0.7505299144	–
20	sym.	–	0.7506549826	0.7506549819
21	sym.	–	0.7507637975	–
22	sym.	–	0.7508611201	–
23	sym.	–	0.7509495679	–
24	sym.	–	0.7510307332	0.7510307324

The published code, composed independently. The authors' ancillary verifier contracts stored intermediate maps with stored decoders, so it does not check that those maps are the channel applied to the encoder. We predicted each of the 330 stored blocks from the stored encoder through our own composition with $\widetilde{\mathcal{N}}$. All agree to 3.8×10^{-14}. Since the basis order inside each irreducible block is not documented, we ran all four orders; the other three miss by at least 6.2×10^{-3}, so the check can fail.

Erasure tolerance. A code built for erasure probability 1/2 should not need it exactly. Holding each certified symmetric code fixed and raising the erasure probability q of A, which keeps A antidegradable and so keeps Lemma 5, the fidelity stays above 3/4 for $q < 0.500400$ ($n = 17$), $q < 0.501296$ ($n = 20$) and $q < 0.502044$ ($n = 24$). These are floating-point values with exact decoders, not certified. The slope $|dF/dq|$ at $q = 1/2$ lies between 0.503 and 0.508 for all three.

VI. TWO USES SUFFICE FROM $d = 20$

Theorem 8 (two uses suffice)**.** *For every $d \geq 20$, $F_c(\widetilde{\mathcal{N}}_p^{\otimes 2}, 2) > 3/4$ at $p = 1/(\sqrt{d}+1)$, and so $F_c((P_d \otimes A_{2d})^{\otimes 2}, 2) > 3/4$.*

Proof. $d \geq 22$ (proven). Take $|0_L\rangle = |++\rangle$ and $|1_L\rangle = |--\rangle$. With no flag Bob inverts the encoder: fidelity 1. With one flag, say on the second use, Bob holds the first qubit intact and a bit c that equals the second qubit's Z value m with probability $1 - p$. Given m, the first qubit

and the reference are in $(|0\rangle_R|+\rangle + (-1)^m|1\rangle_R|-\rangle)/\sqrt{2}$, which X^m maps to the code state; applying X^c is right with probability $1-p$ and otherwise leaves a logical phase error, so the branch has fidelity $1-p$. With two flags, a Z outcome (a,b) leaves the reference in $(|0\rangle + (-1)^{a\oplus b}|1\rangle)/\sqrt{2}$, so Bob learns the reference's X value from the parity of his two bits, which is wrong with probability $2p(1-p)$; preparing the matching X eigenstate gives fidelity $\frac{1}{2} - p(1-p)$. With weights $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$,

$$F = \frac{7 - 6p + 2p^2}{8}. \tag{9}$$

F decreases in p on $[0,1]$, and $F > 3/4$ if and only if $2p^2 - 6p + 1 > 0$, that is $p < (3-\sqrt{7})/2 \approx 0.1771$, that is $\sqrt{d} + 1 > 3 + \sqrt{7}$, that is $d > 11 + 4\sqrt{7} \approx 21.6$. So the code crosses $3/4$ for every $d \geq 22$.

$d = 20, 21$ (certified). The n-use certifier proves $F \geq 0.7508849257$ at $d = 20$ (margin at least 8.8×10^{-4}) and $F \geq 0.7526656475$ at $d = 21$, exactly in $\mathbb{Q}(\sqrt{d})$. For another pair of codes the two-use certifier proves $F \geq 0.7508847632$ and $F \geq 0.7526654055$, and the exact-rational checker, which imports none of our code, reproduces both bounds to the digits shown. The second statement follows by Lemma 4. □

The closed form is also certified directly: $F \geq 0.7509200542$ at $d = 22$, a margin of 9.2×10^{-4}; at $d = 64$, $p = 1/9$ and (9) is exactly $515/648$, above $3/4$ by more than 0.044.

Corollary 9. *For every $d \geq 20$ the channels P_d and A_{2d} exhibit two-shot superactivation at threshold $3/4$, and their onset is exactly $n^* = 2$.*

Proof. Lemma 5 gives the first condition of Definition 1 at $\varepsilon = 1/4$, Theorem 8 the second at $n = 2$, and Lemma 6 excludes $n = 1$. □

What "exactly" covers. Definition 1 depends on ε, and the corollary is for $\varepsilon = 1/4$. For a smaller threshold $1 - \varepsilon < 3/4$ the first condition needs $\sup_m F_c(A_{2d}^{\otimes m}, 2) \leq 1 - \varepsilon$. Nobody has shown that this supremum equals $3/4$; if it is smaller, one use could exceed an admissible threshold below $3/4$, and $n^* = 2$ would not be the least onset over all thresholds.

Better two-use codes. The closed form is not optimal below $d \approx 41.8$. Deforming its two-qubit gate into a controlled phase, $|0_L\rangle = |+\rangle|\beta(\Delta/2)\rangle$ and $|1_L\rangle = |-\rangle|\beta(-\Delta/2)\rangle$ with $|\beta(\phi)\rangle = (|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}$, and optimizing Δ gives

$$F_A = \frac{9 - 2p + \sqrt{\sqrt{d} + (3-2p)^2}}{16} \tag{10}$$

for $d \leq 21 + 12\sqrt{3} \approx 41.8$, where the optimal $\sin(\Delta/2)$ is at most 1; beyond, the family's optimum is the closed form ($\Delta = \pi$), and (10), which then exceeds it, is the fidelity of no code in the family (at $d = 64$ it gives 0.79638 against 515/648). F_A exceeds $3/4$ exactly when $d > (49 - \sqrt{97})/2 \approx 19.6$, equivalently $p < (\sqrt{97} - 1)/48 \approx 0.1843$ (derived; the branch fidelities behind it are matched by our evaluator and by a gate-level simulation). It is certified at $d = 20$, $F \geq 0.7507996270$, and at $d = 32$, $F \geq 0.7685874821$ against the closed form's 0.7679758, a gain of about 6.1×10^{-4}. At $d = 20$ a code with two entangling gates does better (the 0.7508849257 above). At $d = 19$ the best two-use code we found has $F = 0.74903 < 3/4$; with no upper bound on F_c, we do not know whether some two-use code crosses there.

What larger d costs. P_d and A_{2d} act on $\mathbb{C}^{2d}$: one key qubit and a d-dimensional shield. Corollary 9 is about these larger channels. It does not lower the onset for Smith and Yard's $d = 2$ pair, for which Sec. V gives $n^* \leq 15$.

VII. THE ONSET ACROSS THE FAMILY

Table III gives, for $d = 2, \ldots, 21$, the least n at which we found a code above $3/4$. For $n \leq 4$ the codes come from a see-saw over every encoder and decoder in the dense model (4 seeds of 6 restarts each, the best code then polished); for larger n, at $d = 3, \ldots, 6$, from symmetric codes, one search per n (at $d = 5$ run for this revision, by the same search with the same seed, which first reproduced all 28 symmetric entries of the other rows bit for bit), and at $d = 2$ the entry is the two-group code of Theorem 7. Each entry is an upper bound on n^*, found by a local search: the best values of different seeds differ by up to 3.5×10^{-3} (at $d = 12$), and better optimization can only lower the entries. Where marked, the code with its stored decoders is certified exactly in $\mathbb{Q}(\sqrt{d})$ by the n-use certifier, with the thinnest margin, 5.0×10^{-4} ($F \geq 0.7505010800$), at $d = 9$ and $n = 3$; elsewhere the value is a floating-point fidelity, and the entry is an upper bound only as far as that measured value is.

Three facts bear on how to read the table. First, a certified bound below $3/4$ is not a failure of the certifier: at $d = 8$ and $n = 3$ the best code we found has floating-point fidelity 0.7457645, itself below $3/4$, and its certificate gives $F \geq 0.74576$, just below that; the entry at $d = 8$ is therefore the four-use code, and a three-use code we did not find is not excluded. Second, with Lemma 6, $n^* \geq 2$ for every d, so the certified entries bracket the onset: $n^* \in \{2, 3, 4\}$ for $d = 7, 8$; $n^* \in \{2, 3\}$ for $d = 9, \ldots, 19$; and $n^* = 2$ for $d \geq 20$ (Corollary 9). Third, an earlier version of the search used a transposed adjoint of the decoders in its encoder step, so it climbed the wrong functional. A test now checks, for random codes, that the encoder step reproduces the fidelity as a linear functional of the encoder, which the old adjoint fails on every case tried, and the table was recomputed and re-certified after the fix. For $d \leq 6$ no code with $n \leq 4$ that our search found crosses (at $d = 6$, $n = 4$, the best is 0.7477); with the seed spread above, this is not evidence that no such code exists.

TABLE III. The onset across the family, $p = 1/(\sqrt{d}+1)$. $n^* \leq$: the least n at which a code was found above 3/4; F *at* n: that code's fidelity, certified lower bound (truncated) or floating-point value; *best at* $n-1$: the best floating-point value our search found with one use fewer, which bounds nothing about other codes. At $d = 2$ the entry is the two-group code of Theorem 7; symmetric codes first cross at 17. †This code is also certified, $F \geq 0.74576$: a bound just below its floating-point value, which is itself below 3/4 (Sec. VII). ‡Lemma 6.

d	p	$n^* \leq$	F at n	class	best at $n-1$
2	0.4142	15	0.7500784	cert.	0.7497607
3	0.3660	11	0.7509932	float	0.7490092
4	0.3333	9	0.7500942	float	0.7478331
5	0.3090	7	0.7516748	float	0.7488143
6	0.2899	6	0.7540756	float	0.7498886
7	0.2743	4	0.7523282	cert.	0.7405928
8	0.2612	4	0.7565237	cert.	0.7457645†
9	0.2500	3	0.7505010	cert.	0.7258815
10	0.2403	3	0.7548663	cert.	0.7286401
11	0.2317	3	0.7588328	cert.	0.7312636
12	0.2240	3	0.7624592	cert.	0.7337730
13	0.2171	3	0.7657928	cert.	0.7361853
14	0.2109	3	0.7688724	cert.	0.7385142
15	0.2052	3	0.7717298	cert.	0.7407709
16	0.2000	3	0.7743913	cert.	0.7429645
17	0.1952	3	0.7768791	cert.	0.7450819
18	0.1907	3	0.7792121	cert.	0.7471011
19	0.1866	3	0.7814063	cert.	0.7490324
20	0.1827	2	0.7508849	cert.	$\leq 3/4$‡
21	0.1791	2	0.7526656	cert.	$\leq 3/4$‡
≥ 22	≤ 0.1757	2	Eq. (9)	proven	$\leq 3/4$‡

VIII. THE PER-USE REDUCTION IS LOSSY AT ONE USE

Every code above, the published one included, sends each use of $P \otimes A$ through the fixed processing of Lemma 4. For Smith and Yard's pair the joint channel has input dimension 16 and output dimension 20, which Ref. [3] describes as too large to optimize directly. At one use it is not: the erasure flag is classical, so the fidelity splits over the erasure pattern, and a see-saw over every encoder into $\mathbb{C}^{16}$ and every decoder is small.

Theorem 10. *For Smith and Yard's channels,* $F_c(P \otimes A, 2) \geq 0.6627576287 > F_c(\widetilde{\mathcal{N}}, 2) = 1 - \sqrt{2}/4 \approx 0.6464466094$. *One use of the joint channel, coded directly, beats every code on* $\widetilde{\mathcal{N}}$ *by at least 0.0163.*

Proof. The lower bound is certified: an isometric encoder and Gram-factor decoders, positive by construction, normalized and evaluated in ball arithmetic as in Sec. III. *The value of* $F_c(\widetilde{\mathcal{N}}, 2)$ *is proven*: the identity code reaches $\frac{1}{2} + \frac{1-p}{4}$. Conversely $F = \frac{1}{2}F_0 + \frac{1}{2}F_1$ with $F_0 \leq 1$. In the flagged branch the composite channel is measure-and-prepare, $\rho \mapsto \sum_y \mathrm{Tr}(N_y\rho)\,\sigma_y$, with $N_y = \sum_x T(y|x)M_x$ for a POVM $\{M_x\}$ on the encoded qubit, T the binary symmetric channel with flip probability p, and states σ_y. Its fidelity with Φ is $\frac{1}{4}\sum_y \mathrm{Tr}(N_y\sigma_y) \leq \frac{1}{4}\sum_y \|N_y\|$. Writing $N_0 = p\mathbb{1} + (1-2p)M_0$ and $N_1 = (1-p)\mathbb{1} - (1-2p)M_0$ gives, since $0 \leq M_0 \leq \mathbb{1}$ and $p \leq 1/2$ here, $\|N_0\| + \|N_1\| \leq 1 + (1-2p) = 2-2p$, so $F_1 \leq (1-p)/2$ and $F \leq \frac{1}{2} + \frac{1-p}{4}$. At $p = \sqrt{2}-1$ this is $1 - \sqrt{2}/4$. □

The reduction's decoder is optimal for its encoder (measured: lifted through its encoder, the identity code at one use and the best two-use $\widetilde{\mathcal{N}}$ code we know score their $\widetilde{\mathcal{N}}$ values in the direct model; a test holds the first); the loss is in the encoder, which the direct code changes to give up perfect recovery when A delivers in exchange for more when it erases. Only the one-use statement is shown; we have no certified or committed direct code at two uses.

IX. WHAT IS NOT SHOWN: LOWER BOUNDS ON THE ONSET

Every entry of Table III is an upper bound. A lower bound $n^* > m$ needs an upper bound of at most 3/4 on F_c of m joint uses, and the only one we have is Lemma 6, for $m = 1$.

A PPT relaxation cannot do it. The standard computable upper bound on $F_c(\mathcal{N}, M)$ is the PPT-assisted channel fidelity of Leung and Matthews [12], the semidefinite program

$$\begin{aligned} F_{\mathrm{PPT}}(\mathcal{N}, M) = \max\ \mathrm{Tr}(J_{\mathcal{N}}W) \text{ subject to } & \mathrm{Tr}\,\rho = 1, \\ & 0 \leq W \leq \rho \otimes \mathbb{1}_B, \\ & -\rho \otimes \mathbb{1}_B \leq MW^{T_B} \leq \rho \otimes \mathbb{1}_B, \end{aligned}$$

with $J_{\mathcal{N}} = \sum_{ij} |i\rangle\langle j| \otimes \mathcal{N}(|i\rangle\langle j|)$, so that $\mathrm{Tr}_B\, J_{\mathcal{N}} = \mathbb{1}_A$; here $M = 2$. (Leung and Matthews write the objective with $J_{\mathcal{N}}^{\mathsf{T}}$; the value is the same, because $(W, \rho) \mapsto (W^{\mathsf{T}}, \rho^{\mathsf{T}})$ maps the feasible set onto itself and $\mathrm{Tr}(J_{\mathcal{N}}^{\mathsf{T}}W^{\mathsf{T}}) = \mathrm{Tr}(J_{\mathcal{N}}W)$.)

Proposition 11. *For every* n *and every* $p \in [0,1]$, $F_{\mathrm{PPT}}(\widetilde{\mathcal{N}}_p^{\otimes n}, 2) = F_{\mathrm{PPT}}(A^{\otimes n}, 2) = 1 - 2^{-(n+1)}$, *where* A *is the 50% erasure channel on a qubit. For any channel* P *and any 50% erasure channel* A' *on* $\mathbb{C}^D$, $D \geq 2$, $F_{\mathrm{PPT}}((P \otimes A')^{\otimes n}, 2) \geq 1 - 2^{-(n+1)}$, *with equality when* P *is PPT.*

Proof. In each case the erasure pattern $z \in \{0,1\}^n$ is classical at the output (for A and A' the flag is the erasure symbol's orthogonal sector): the channel is $\sum_z 2^{-n}\mathcal{N}_z \otimes |z\rangle\langle z|$, where the channel $\mathcal{N}_z$ acts on the kept uses as the identity (tensored with P for the joint channel) and on the erased ones as $\mathcal{X}^p \circ \overline{\Delta}$, as preparation of the erasure symbol, or as P tensored with that preparation. So $J = \bigoplus_z 2^{-n} J_z$ with $\mathrm{Tr}_B\, J_z = \mathbb{1}$.

Upper bound. Let $J_e = 2^{-n}J_{1\cdots1}$ be the all-erased block. Its channel is PPT: an entanglement-breaking map is, and so is $P^{\otimes n}$ tensored with a preparation when P is PPT. Hence $Y = J - J_e \geq 0$ and $Z = J_e^{T_B}/M \geq 0$,

and for every feasible (W, ρ), by $W \le \rho \otimes \mathbb{1}$ and $MW^{T_B} \le \rho \otimes \mathbb{1}$,

$$\begin{aligned}\operatorname{Tr}(JW) &= \operatorname{Tr}(YW) + M \operatorname{Tr}(ZW^{T_B}) \\ &\le \operatorname{Tr}\big((Y+Z)(\rho\otimes\mathbb{1})\big) = 1 - 2^{-n} + 2^{-n}/M,\end{aligned}$$

since $\operatorname{Tr}_B(Y+Z) = (1-2^{-n}+2^{-n}/M)\mathbb{1}$. This is a dual certificate, with no assumption on the form of W; at $M=2$ it is $1-2^{-(n+1)}$.

Feasible point. Let $\rho = \mathbb{1}/2^n$ and $W = \bigoplus_z W_z$, block diagonal in z; the blocks are labelled by basis states, so the partial transpose keeps them and each constraint holds if it holds on every block. For a pattern whose first kept use is i, let $W_z = \frac{1}{2}\Phi_{A_iB_i} \otimes \rho_{\text{rest}} \otimes \mathbb{1}_{\text{rest}}$, with Φ the projector on $|\Phi\rangle$ and $\rho_{\text{rest}} = \mathbb{1}/2^{n-1}$. Then $0 \le W_z \le \rho \otimes \mathbb{1}$; $MW_z^{T_B} = \frac{1}{2}F_i \otimes \rho_{\text{rest}} \otimes \mathbb{1}$ with F_i the swap, whose eigenvalues are ± 1, lies between $-\rho\otimes\mathbb{1}$ and $\rho\otimes\mathbb{1}$; and $\operatorname{Tr}(J_z W_z) = \operatorname{Tr}(J_{\text{id}}\frac{1}{2}\Phi)\operatorname{Tr}(J_{\text{rest}}(\rho_{\text{rest}}\otimes\mathbb{1})) = 1$. For the all-erased pattern, $W_z = \frac{1}{2}\rho\otimes\mathbb{1}$ is feasible with value $\frac{1}{2}$. The value, $1-2^{-n}+2^{-n}/2$, meets the upper bound.

The joint channel. Embed the qubit in $\mathbb{C}^D$ by a real isometry, which keeps every constraint, and let P's inputs be $|0\rangle\langle 0|$ and its outputs be traced: $W' = W \otimes \bigotimes_i (|0\rangle\langle 0| \otimes \mathbb{1})$, $\rho' = \rho \otimes \bigotimes_i |0\rangle\langle 0|$ is feasible for $(P \otimes A')^{\otimes n}$, with the same value because $\operatorname{Tr}(J_P(|0\rangle\langle 0| \otimes \mathbb{1})) = 1$. When P is PPT the upper bound applies. □

Every P_d at its PPT point $p_1 = 1/(\sqrt{d}+1)$, the one used throughout, is PPT (Theorem 3), so for every pair of this paper the relaxation's value at two joint uses is exactly $7/8$, and since a lower bound $n^* > 2$ needs an upper bound of at most $3/4$ there, it can never certify one. The proposition is also checked, and each check covers only what it states. In exact arithmetic on the full output space, with the erasure symbol a one-dimensional sector and no block structure assumed, the feasible point and the dual certificate of the proof are verified at $n = 1, \dots, 3$ for $\widetilde{\mathcal{N}}_p$ at each of 6 rational values of $p \in [0,1]$ and for A, and for Smith and Yard's joint channel $P_2 \otimes A_4$ at $n=1$ (dimension 320, in $\mathbb{Q}(\sqrt{2})$). Each of 13 controls fails as it must: a block without its factor $\frac{1}{2}$ or scaled by $\frac{1}{2}+\frac{1}{64}$, the dual with $t-\frac{1}{64}$, and the dual of two channels whose all-erased block is not PPT. A dense floating-point solution of the program, which builds one block per flag pattern with a common ρ and so presupposes the decomposition rather than testing it, gives $3/4$ and $7/8$ for both channels at one value of p (to 1.1×10^{-8}). PPT-assisted codes do not respect the two-extendibility that caps the erasure channel. A relaxation that keeps it, such as the extendibility hierarchies for bilinear problems of Berta *et al.* [13] with permutation symmetry, is a candidate; we have not attempted it, and its size at the n that matters here is unestimated.

What is known. At threshold $3/4$: $n^* = 2$ for $d \ge 20$; $2 \le n^* \le 3$ for $d = 9, \dots, 19$; $2 \le n^* \le 4$ for $d = 7, 8$; and $2 \le n^* \le 15$ for Smith and Yard's pair. These statements are for that threshold and do not carry over to another. Above it, every threshold $1-\varepsilon \in (3/4, 1)$ is admissible, since Lemma 5 bounds each channel alone by $3/4$, but the onset can only grow: $F_c \le F_{\text{PPT}}$, so Proposition 11 gives $F_c((P_d \otimes A_{2d})^{\otimes n}, 2) \le 1 - 2^{-(n+1)}$ for every pair of this paper, and n uses can exceed $1-\varepsilon$ only if $2^{-(n+1)} < \varepsilon$. Two uses therefore exceed no threshold of $1-2^{-3} = 7/8$ or more. (Directly: both erasures occur with probability $1/4$, and in that branch the code acts through the PPT channel $P_d^{\otimes 2}$ alone, so its fidelity there is at most $1/2$.) Below $3/4$ the definition first needs $\sup_m F_c(A_{2d}^{\otimes m}, 2) \le 1-\varepsilon$, and whether that supremum equals $3/4$ is open.

X. OUTLOOK: HARDWARE

For context only, and demonstrating nothing: one preregistered run on `ibm_kingston` at $d = 32$, whose circuits replaced an erased use by its classical equivalent, measured an entanglement fidelity of 0.615 ± 0.009, 4.6 standard errors below its model's prediction of 0.657, and simulated trapped-ion devices built from published specifications clear $3/4$ by projected margins of only $+0.016$ to $+0.024$ at $d = 64$, less than that model's miss. The run, the models and a frozen, unrun trapped-ion package of 8,000 circuits at $d = 256$ are in `evidence/superactivation/` (`hw_run_m5/`, `ion_run_m6/`, `ion_run_m8_r224/`).

XI. RELATED WORK

Superactivation of quantum capacity is due to Smith and Yard [1], building on bound entanglement [14] and private states [15]. It was extended to optical channels with attenuation and Gaussian noise by Smith, Smolin and Yard [16], and shown by Brandão, Oppenheim and Strelchuk to occur for a wide class of inequivalent channels, including one built from the depolarizing channel [17]. Its non-asymptotic form, the definition we use, Imported results 1 and 2 and the general effective-channel theorem of their supplement are those of Parentin, Bergh, Datta and Wilde [3], with the permutation-invariant framework of Bergh and Parentin [4]. Cubitt *et al.* showed that detecting quantum capacity may require an unbounded number of uses [2]. Superactivation of zero-error capacities is a different notion, shown already at a single use [18] and for the asymptotic zero-error classical capacity [19]. Wu and Wang study capacity amplification through private channels whose Choi operators are private states, and give an alternative proof of superactivation that extends it to a broader parameter regime [20]. In its version of 5 October 2026, Zhu and Wang's work [21] shows that a fixed qutrit channel superactivates every finite-dimensional channel with zero quantum but positive private capacity. These works concern capacities: the statements are asymptotic, the last one existential, and we found no number of uses in any of them. Activating private capacity [22, 23] is a different notion again. Khanian, Leung and Smith re-

late permutation-invariant inputs to finite-block coherent information, without a superactivation result [24]. An NMR experiment reports superactivation by coherently superposing channels [25], a different mechanism from the parallel uses here.

Novelty. In a literature search logged on 3 November 2026, with earlier searches on 13 and 14 October and an addendum on 3 November, we found no certified onset below 17 for Smith and Yard's pair, no superactivation of quantum capacity in two uses in the sense of Definition 1, and no study of the onset as a function of shield dimension. The searches were not exhaustive, and we make no priority claim beyond this.

XII. SCOPE AND LIMITATIONS

What the proofs cite. The proofs use two standard fidelity bounds, for PPT states [7] and for two-extendible states [8, 9], and data processing. Imported results 1 and 2 are the $d = 2$ instances of Lemmas 5 and 4; no statement depends on them.

Reductions inside the certifiers. The symmetric and two-group certifiers compute in the reduced representation of Lemma 2, which is proven; their own ball-arithmetic matrices and coefficients are checked against the dense construction at every size a certificate uses, and the certifiers themselves are re-run end to end on every witness except the two-group ones at $n = 15, 16$, whose certificates take hours (Sec. III). The n-use, two-use and exact-rational certifiers work in the dense model.

Upper bounds only. Apart from Lemma 6, nothing here bounds n^* from below, and the searches behind Table III are local.

Floating point. The symmetric codes between certificates, the onset entries marked float, the erasure tolerances, the best code at $d = 19$, family A's formula, and the hardware numbers are not certified.

Different channels. Two-shot superactivation for $d \geq 20$ concerns larger channels than Smith and Yard's (Sec. VI).

XIII. REPRODUCIBILITY

Every path in this paper is relative to the subdirectory `superactivation/` of release v2026.11.07.2 of `https://github.com/coherence-energy-labs/oneq-certificates`, a bundle signed and time-stamped separately from the rest of that repository, which does not contain these paths. The development history behind it is private and is available to referees on request. The code is in `experiments/superactivation/` (`shield_family/` for the family, the onset search, the n-use and two-use certifiers and `p2_kraus_comparison.py`; `general_codes/` for the two-group codes, their certificate and `verify_reduction_dense.py`; `direct/` for Sec. VIII; `ppt_relaxation_probe.py`, `ppt_relaxation_fullspace.py`, `erasure_sensitivity.py` and `stored_maps_entrywise.py`), in `src/oneq/superactivation.py` and `superactivation_certificate.py`, and in the exact-rational checker `tools/verify_r142_rational_witness.py`. The evidence is in `evidence/superactivation/` (`general_codes/`, `shield_family/`, `direct/` and the certificate, curve, erasure, stored-map and PPT-probe records at its top level); the dense check, the comparison of P_2 and the full-space check of Proposition 11 are `general_codes/reduction_dense_check_v2.json`, `shield_family/p2_kraus_comparison_v2.json` and `ppt_relaxation_fullspace.json` (each `_v2` supersedes the file of the same name without the suffix, and the last supersedes `ppt_relaxation_exact.json`, which assumed the block form). The two-group certificates take hours of exact arithmetic; the tests bind them to their stored bytes and recompute the stored pairs, the dense check of Sec. III recomputes them outside the reduction (about 15 min), and every n-use certificate re-derives in under a second. The stored-map check of Sec. V needs the authors' arXiv source archive, which is not redistributed.

Stored decoders. An n-use certificate's decoder file holds, for each flag pattern z, a factor K_z with 2^{n+1} rows. Reshaping a column of K_z to a 2×2^n matrix, whose row is the reference qubit (the decoder's output) and whose column indexes the n received qubits (the first most significant), and taking its complex conjugate gives a Kraus operator of the decoding channel. Equivalently, $D_z = K_z K_z^\dagger$ is the Choi matrix of the decoder's adjoint map with the output index first, $\mathrm{Tr}_R\, D_z = \mathbb{1}$, and the fidelity of pattern z is $\mathrm{Tr}(\sigma_z D_z)/4$ for Bob's state $\sigma_z = (\mathrm{id}_R \otimes \mathcal{N}_z)(|w\rangle\langle w|)$, $|w\rangle = \sum_r |r\rangle \otimes V|r\rangle$; only this reading reproduces the branch values the certificates record. The symmetric and two-group decoders are stored in the same convention on the Dicke registers of Lemma 2 (the two-group files keep the upper triangle of each D).

Every number in this paper is a macro written by `paper/tools/make_numbers_superactivation.py` from those committed files; it re-proves every n-use certificate and the exact-rational bounds, recomputes the closed forms and their thresholds, and refuses to run when a certificate, or the record of the $d = 5$ symmetric search, no longer binds to its bytes, when the dense check, the comparison of P_2 or the full-space check of Proposition 11 was produced by other code than the tree holds, or when the evidence no longer supports a comparison it checks; it checks every comparison the text draws between numbers from those files. Not checked by it: statements about the literature, which rest on reading the sources, and those the text attributes to a test (the spectrum of Theorem 3 and the match with the state of Ref. [5], the isometric encoders

of Sec. V, the encoder step of Sec. VII, the decoder optimality of Sec. VIII, and family A's branch fidelities), which the test suite holds. Each number is rounded so that the sentence quoting it stays true: certified lower bounds, margins and every "at least" are truncated, every "less than", "at most" or "up to" is rounded up, ranges are enclosed outward, and floating-point values are rounded to nearest, never onto or across 3/4.

Use of AI tools. AI tools assisted with software development, the proofs and manuscript preparation. The AI-agent checks of the two-group certificate, of the family's argument and of Proposition 11 (whose exact full-space check began as one), and two AI-agent reviews of earlier drafts of this manuscript, were each carried out by an agent that had not written the artifact under review. Two external audits of this manuscript's first release are archived with it (`docs/reviews/`): the first calls itself AI-assisted, and the second states that it was done by an AI model with AI sub-agents. None of these is third-party peer review. The author is responsible for all content.

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